

Full Stack Development with MERN

Project Documentation format

1. Introduction

- **Project Title:** House Rent
- **Team Members:** N. Chaithanya Krishna – Project Development, Documentation, Testing & Presentation

2. Project Overview

- **Purpose:** The House Rent project is a web-based rental management system designed to simplify the process of renting residential properties. It provides a centralized platform where renters can search and book verified properties, owners can manage property listings and bookings, and administrators can oversee the entire system efficiently.
- **Features:**
 - User Registration & Login
 - Role-Based Authentication (Admin, Owner, Renter)
 - Property Upload & Management
 - Property Search & Filters
 - Property Booking
 - Booking Approval / Rejection
 - Admin Dashboard
 - Responsive UI
 - Dark/Light Mode

3. Architecture

- **Frontend:** The frontend is developed using React.js, HTML5, CSS3, JavaScript, and Bootstrap to create a responsive and user-friendly interface.
- **Backend:** The backend is developed using Node.js and Express.js, providing RESTful APIs, authentication, booking management, and business logic.
- **Database:** MongoDB stores user information, property details, booking records, and transaction history.

4. Setup Instructions

- **Prerequisites:**
 - Node.js
 - npm
 - MongoDB
 - Visual Studio Code

- **Installation:**
git clone <repository-url>
cd frontend
npm install
cd ../backend
npm install
Create the required `.env` file and configure the MongoDB connection.

5. Folder Structure

- **Frontend:**

frontend/

├── src/

├── modules/

├── components/

├── App.js

└── package.json

- **Backend:**

backend/

├── routes/

├── models/

├── controllers/

├── middleware/

├── index.js

└── package.json

6. Running the Application

- Provide commands to start the frontend and backend servers locally.
 - **Frontend:**

```
cd frontend
npm start
```

- **Backend:**

```
cd backend
npm start
```

7. API Documentation

Method	Endpoint	Description
POST	/register	Register User
POST	/login	Login User
GET	/properties	View Properties
POST	/property	Add Property

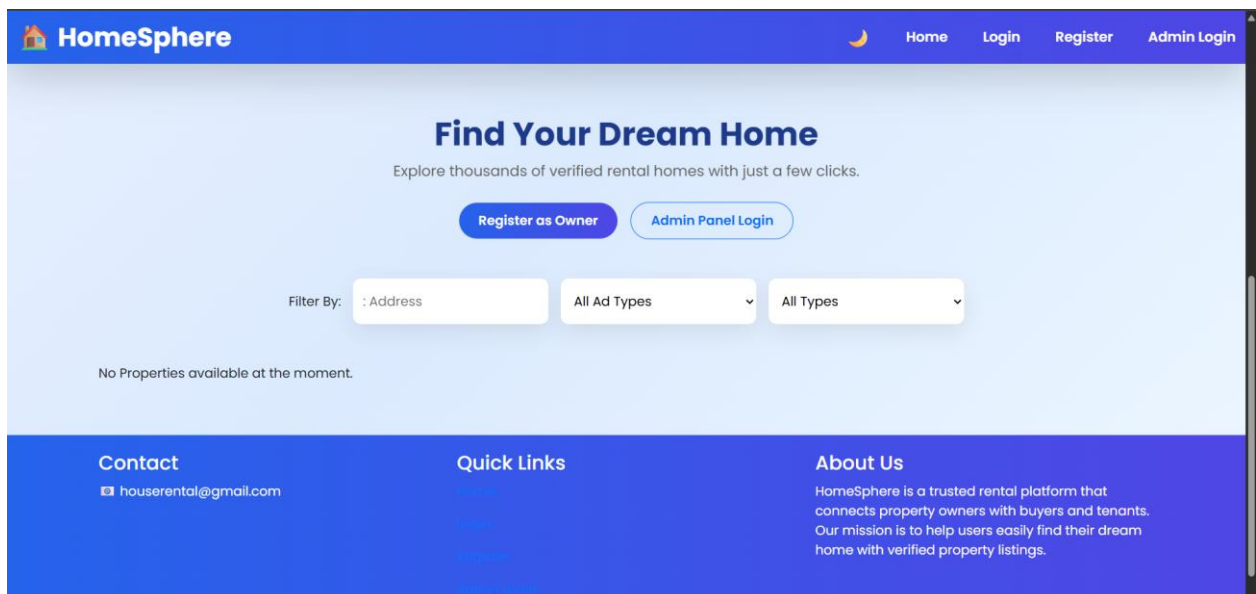
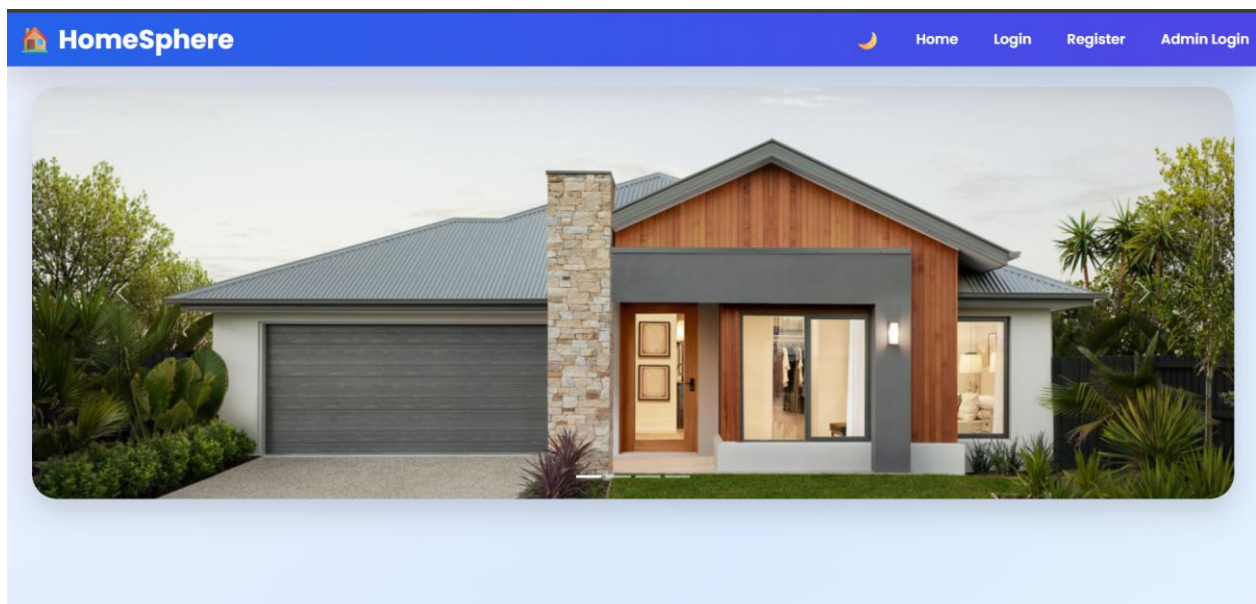
Method	Endpoint	Description
PUT	/property/:id	Update Property
DELETE	/property/:id	Delete Property
POST	/booking	Book Property
GET	/bookings	View Bookings

8. Authentication

- Authentication is implemented using **JWT (JSON Web Token)**. Passwords are encrypted using **bcrypt.js**, and role-based authorization is provided for Admin, Owner, and Renter.

9. User Interface

Home Page:



Register or Sign Up:

 HomeSphere

HomeLoginRegister


Sign Up

SIGN UP

Already have an account? [Sign In](#)

Login:

 HomeSphere

HomeLoginRegister

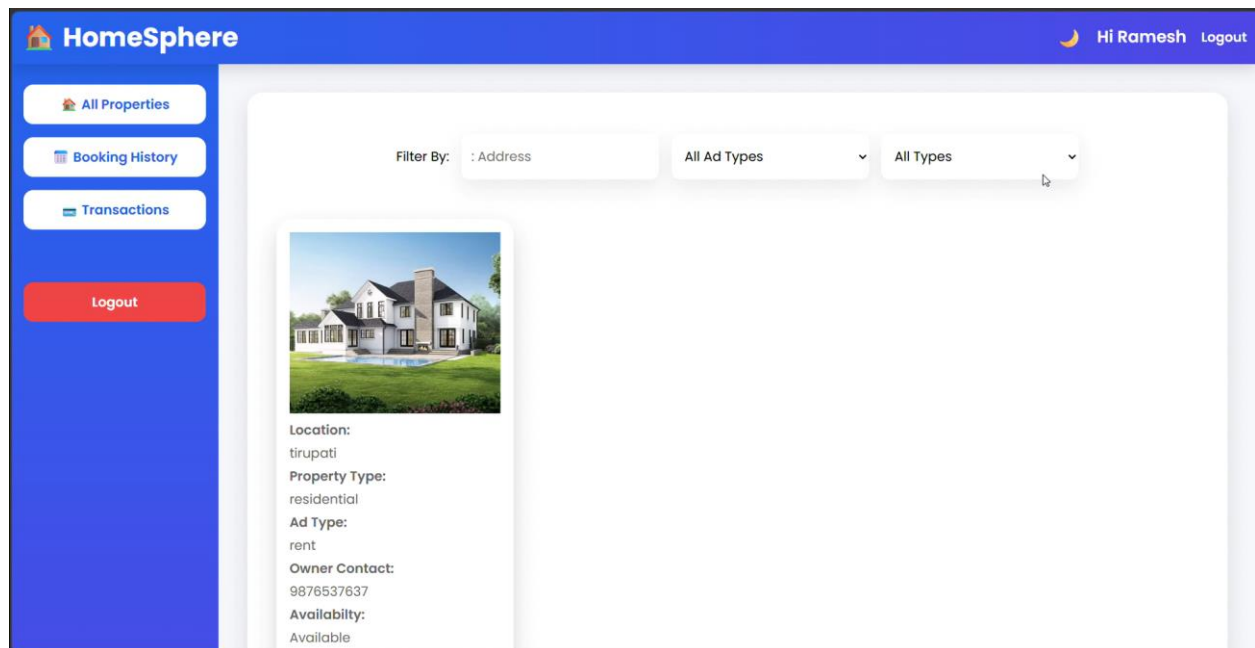

Sign In

SIGN IN

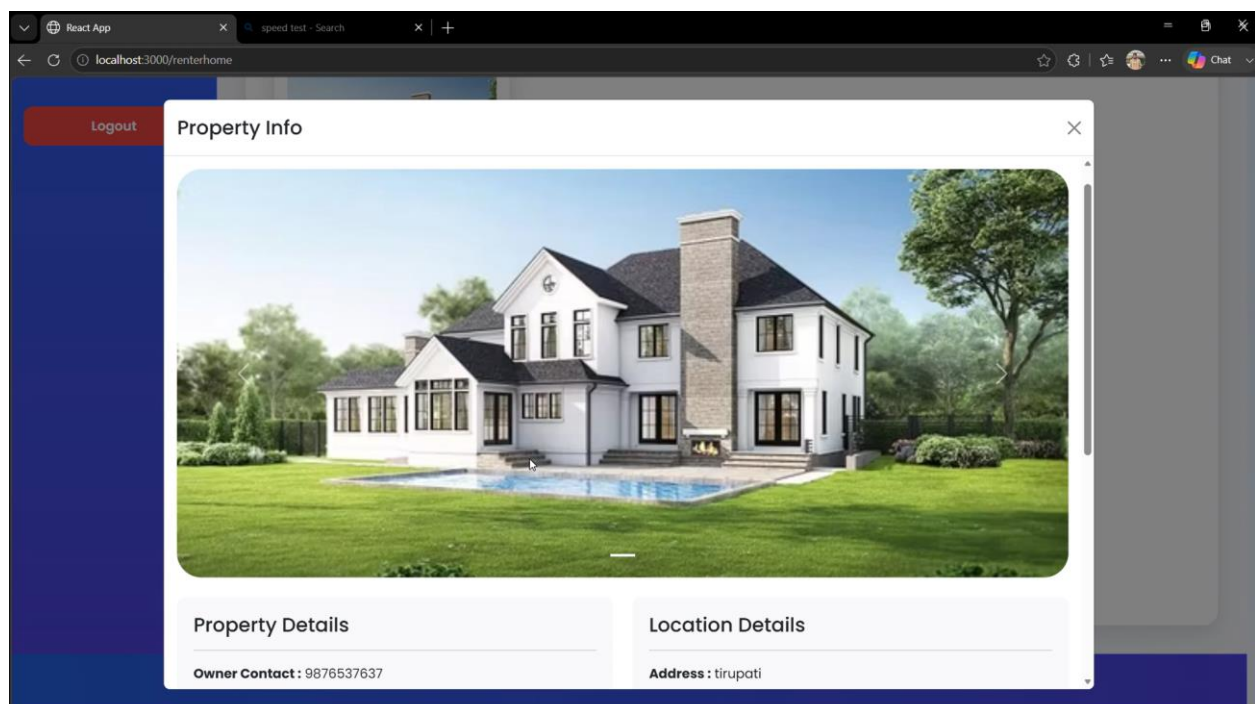
Forgot Password? [Click here](#)

Don't have an account? [Sign Up](#)

Renter Dashboard:



Booking Page:



[Logout](#)

Property Info

Property Details

Owner Contact : 9876537637

Availability : Available

Property Amount : ₹20000

Location Details

Address : tirupati

Property Type : residential

Ad Type : rent

Additional Information

tirupati

Additional Info: tirupati

Confirm Your Booking

Full Name

Phone Number

Book Property

Booking History:

HomeSphere

Hi Ramesh [Logout](#)

All Properties

Booking History

Transactions

Logout

Booking ID	Property ID	Tenant Name	Phone	Booking Status	Actions
6a5669302d19f3887efce3db	6a5668ab2d19f3887efce3cb	Ramesh	987654321	pending	

Admin Login:

Admin Login

Sign in with your admin credentials

Login

Admin Dashboard:

HomeSphere

All Users

All Properties

All Bookings

Logout

Admin Dashboard

Hi Admin

User ID	Name	Email	Type	Granted (Old Logic)	isApproved (New)	Actions
6a4348a781b44fb517f920c3	Admin	chaituyadav03@gmail.com	Admin	granted	☒	TOGGLE GRANTED UNAPPROVE DELETE
6a567d74417beacd6efb5db	Maresh	Maresh@gmail.com	Owner	ungranted	☒	TOGGLE GRANTED UNAPPROVE DELETE

Owner Dashboard:

HomeSphere

Hi Maresh

Logout

+ Add Property

All Properties

Bookings

Transactions

Logout

Property type

Property Ad type

Property Full Address

Residential

Rent

Address

Property Images

Owner Contact No.

Property Amt.

Choose Files

No file chosen

contact number

0

Additional details for the Property

Submit form

Owner Bookings:

HomeSphere

Hi Maresh

Logout

+ Add Property

All Properties

Bookings

Transactions

Logout

Booking ID	Property ID	Tenant Name	Tenant Phone	Booking Status	Actions
6a5669302d19f3887efce3db	6a5668ab2d19f3887efce3cb	Ramesh	987654321	pending	Change

10. Testing

- The application was tested using manual testing to verify user registration, login, property management, booking workflow, transaction history, authentication, and responsive design.

11. Screenshots or Demo

<https://drive.google.com/file/d/1r8i3LsaqftZga0lEBF10kHr0sMdpK7tG/view>

12. Known Issues

- Currently, no major issues were identified during testing. Minor UI enhancements and additional features may be implemented in future versions

13. Future Enhancements

- Online Payment Gateway
- Google Maps Integration
- AI-Based Property Recommendation
- Chatbot Support
- Email & SMS Notifications
- Wishlist Feature
- Mobile Application
- Property Rating & Reviews

